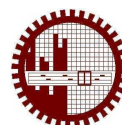


**Department
of Computer
Science and
Engineering,
BUET**



Description of Course CSE 220

PART A: General Information

- 1 **Course Title** : SIGNALS AND LINEAR SYSTEMS
SESSIONAL
- 2 **Type of Course** : SESSIONAL
- 3 **Offered to** : DEPARTMENT OF CSE
- 4 **Pre-requisite Course(s)** : None

PART B: Course Details

1. **Course Content (As approved by the Academic Council)**

Sessional based on CSE 219.

2. **Course Objectives**

The students are expected to:

- i. Learn continuous and discrete-time signals, linear systems and their properties.
- ii. Learn Fourier, DFT, Laplace and Z transforms
- iii. Apply the principles of Fourier DFT, Laplace and Z transforms for analyzing signals and linear systems.

3. **Knowledge required**

Technical

- Basic of calculus and differential equations

4. **Course Outcomes (COs)**

CO No.	CO Statement After undergoing this course, students should be able to:	Corresponding PO(s)*	Domains and Taxonomy level(s)**	Delivery Method(s) and Activity(-ies)	Assessment Tool(s)
CO1	Understand continuous and discrete-time signals, linear	-	C2	Lecture, Demonstration, and hands-on	Assignments and Quiz



	systems, their properties, and the sampling process					
CO2	Apply mathematical concepts and tools for analysis of signals and linear systems	PO1, PO5	C3	Lecture, Demonstration, and hands-on	Assignments and Quiz	
CO3	Analyze continuous and discrete-time signals, and linear systems in time and frequency domains	PO1, PO2	C4	Lecture, Demonstration, and hands-on	Assignments and Quiz	

***Program Outcomes (POs)**

PO1: Engineering knowledge; PO2: Problem analysis; PO3: Design/development of solutions; PO4: Investigation; PO5: Modern tool usage; PO6: The engineer and society; PO7: Environment and sustainability; PO8: Ethics; PO9: Individual work and teamwork; PO10: Communication; PO11: Project management and finance; PO12: Life-long learning.

****Domains**

C-Cognitive: C1: Knowledge; C2: Comprehension; C3: Application; C4: Analysis; C5: Synthesis; C6: Evaluation

A-Affective: A1: Receiving; A2: Responding; A3: Valuing; A4: Organizing; A5: Characterizing

P-Psychomotor: P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Organization

5. Mapping of Knowledge Profile, Complex Engineering Problem Solving and Complex Engineering Activities

COs	K1	K2	K3	K4	K5	K6	K7	K8	P1	P2	P3	P4	P5	P6	P7	A1	A2	A3	A4	A5
CO1		√	√	√	√	√			√	√	√	√				√			√	
CO2		√	√	√	√	√			√	√	√	√				√	√			
CO3			√	√	√	√		√	√	√	√				√	√		√		

K-Knowledge Profile:

K1: A systematic, theory-based understanding of the natural sciences applicable to the discipline; **K2:** Conceptually based mathematics, numerical analysis, statistics and the formal aspects of computer and information science to support analysis and modeling applicable to the discipline; **K3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline; **K4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline; **K5:** Knowledge that supports engineering design in a practice area; **K6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline; **K7:** Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the engineer's professional responsibility to public safety; the impacts of engineering activity; economic, social, cultural, environmental and sustainability; **K8:** Engagement with selected knowledge in the research literature of the discipline



require abstract thinking, originality in analysis to formulate suitable models; **P4:** Involve infrequently encountered issues; **P5:** Are outside problems encompassed by standards and codes of practice for professional engineering; **P6:** Involve diverse groups of stakeholders with widely varying needs; **P7:** Are high level problems including many component parts or sub-problems

A-Range of Complex Engineering Activities:

A1: Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information and technologies); **A2:** Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering or other issues; **A3:** Involve creative use of engineering principles and research-based knowledge in novel ways; **A4:** Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation; **A5:** Can extend beyond previous experiences by applying principles-based approaches

6. Lecture/ Activity Plan

Week	Lecture Topics	Corresponding CO(s)
1	Lecture 1: Python Basics Introduction, Installation, Tools, Basic Python: variables, user input, branching, loops, functions etc.	CO1
2	Lecture 2: Python Advanced Numpy, matplotlib, plotting basic signals	CO1
3	Practice Problems on Python Hands on practice on the topics covered in lecture 1 and 2	CO1, CO2
4	Online 1: Signals and their Properties Causality, Linearity, Time Invariance, Time/Phase Shift, Reversal etc.	CO1, CO2
5	Practice Problems on Convolution Hands on practice problems on Continuous and Discrete Convolution Offline 1: Convolution Continuous and Discrete Convolution Implementation and Applications	CO2
6	Online 2: Convolution Convolution problems building on offline 1 Offline 1 Evaluation	CO2
7	Offline 2: Fourier Series & CFT Implementation and Application of Fourier Series and CFT Project Assignment & Group Formation	CO2, CO3
8	Online 3: Fourier Series & CFT Problems on Fourier Series and CFT building on offline 2 Offline 2 Evaluation	CO2, CO3
9	Offline 3: DFT & FFT Implementation and Application of DFT and FFT	CO2, CO3
10	Online 4: DFT & FFT Problems on DFT and FFT building on offline 3 Offline 3 Evaluation	CO2, CO3
11	Project Update	CO2
12	<i>Reserved</i>	
13	Online 5: Sampling, Filtering, Laplace & Z-transform Problems on sampling/filtering/Laplace/Z-transform	CO1, CO2, CO3,
14	Quiz	CO1, CO2, CO3



every class.

- Assignments: There will be some online and offline assignments, as well as a group based project assignment.
- Quiz Exam: A comprehensive Quiz exam will be held at the end of the semester.

8. Distribution of Marks (tentative)

Attendance:	10%
Assignments:	70%
Quiz Exam:	20%
Total:	100%

9. Textbook/ Reference

- a. Oppenheim, Alan V., and A. S. Willsky.
Signals and Systems.

COURSE OUTLINE

Course No: CSE 220, Level 2/
Term 2, Credit (Contact) Hours:
0.75 Credits (1.5hrs/wk)

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